

Data Journalism

Session 5: Sampling and surveys

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Hertie School | GRAD-E1493

Today

1. Surveys in the wild
2. How surveys are made
3. Survey sampling 101
4. Take-aways

Surveys in the wild

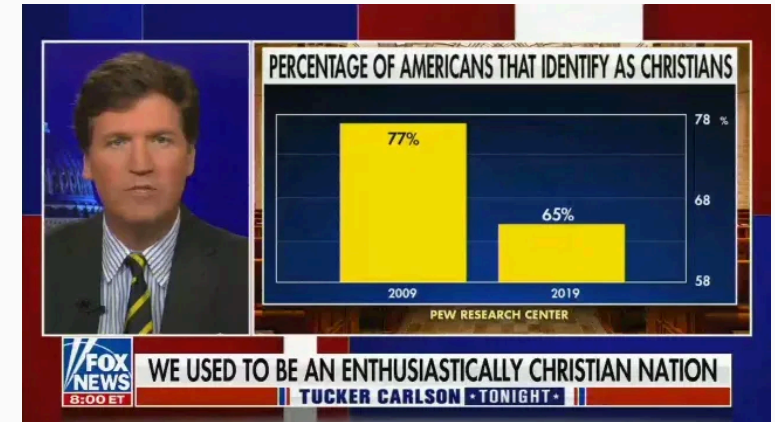
Why should data journalists care about surveys?

Surveys are everywhere in the news: election polls, consumer confidence, trust in institutions, vaccination rates, public opinion on any policy issue imaginable.

As a data journalist, you will rarely design your own survey — but you will **consume, evaluate, and report on** surveys almost daily.

Key questions

- How was this survey conducted?
- Who was sampled and how?
- Does the survey design (not just sample size) justify the claims?
- What was the exact question wording?
- What is the (real) margin of error?



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← Article ↻

Elon Musk's Poll on the Formation of a New Party Attracts 64% of Votes

blogtienso Jul 5, 2025

The United States, home to the most billionaires, will celebrate its 249th Independence Day on July 4, 2025.

However, Elon Musk, the richest man in the world, who has previously supported cryptocurrencies like Bitcoin and Dogecoin, is dissatisfied with the two-party political system in this country.

Musk, who until recently supported President Donald Trump, asked on X this July 4 whether the United States should establish a new political party.

Exercise: How would you report this?

Scenario A

A press release from a political consulting firm says:

"72% of Germans support a ban on combustion engine cars by 2035."

The survey was conducted online; n = 1,200

Discuss (3 min)

- What questions would you ask before reporting this number?
- What information is missing?

Exercise: How would you report this?

Scenario A

A press release from a political consulting firm says:

"72% of Germans support a ban on combustion engine cars by 2035."

The survey was conducted online; n = 1,200

Discuss (3 min)

- What questions would you ask before reporting this number?
- What information is missing?

Scenario B

A new study in *The Lancet* reports:

"Vaccine hesitancy increased by 12 percentage points in 6 months, based on a sample of 250,000 respondents recruited via Facebook Ads."

Discuss (3 min)

- What do you think about the sample size?
- How would you report the uncertainty around the "12 percentage points"?

International guidelines for reporting on polls

AAPOR (USA)

The **Journalist Cheat Sheet** (aapor.org) distills the essentials: Who paid? Probability or non-probability sample? Exact question wording? Margin of error? What do other polls say? Interesting rule: **"Error margin should not be reported for non-probability samples."**

ESOMAR/WAPOR

The **Guideline on Opinion Polls and Published Surveys** (esomar.org) is the international reference. Their 2023 Freedom Report found that experts rate the quality of journalistic poll reporting as low in more than half of surveyed countries.

Pew Research Center

Tips for Journalists Covering Polls (2024; pewresearch.org), originally published in the IRE Journal. Common mistakes: treating MoE as total error; reporting decimal places; calling 51% a "majority" when MoE is ± 3 . **Check the link for a collection of very useful advice.**



Pew Research Center

German guidelines (for those who are interested)

Germany

Pressekodex, Richtlinie 2.1 ([presserat.de](https://www.presserat.de)): When publishing poll results, the press must report the number of respondents; the timing of fieldwork; the exact question wording; the commissioning party; and whether results are representative. Non-representative online polls must be labeled as such (Presserat ruling, March 2018).

Richtlinie für die Veröffentlichung von Ergebnissen der Markt- und Sozialforschung (2024/2025; [rat-marktforschung.de](https://www.rat-marktforschung.de)): Binding industry code by ADM, ASI, BVM, DGOF. Requires institute name, client, population, sample size, fieldwork dates, mode, weighting, and question wording for every published study. Applies via *Verkehrssitte* even to non-members.

A test case: Civey vs. Presserat

In 2018, Forsa, Infas, and Forschungsgruppe Wahlen complained to the Presserat that Focus Online had published a Civey poll as "representative" without adequate scrutiny.

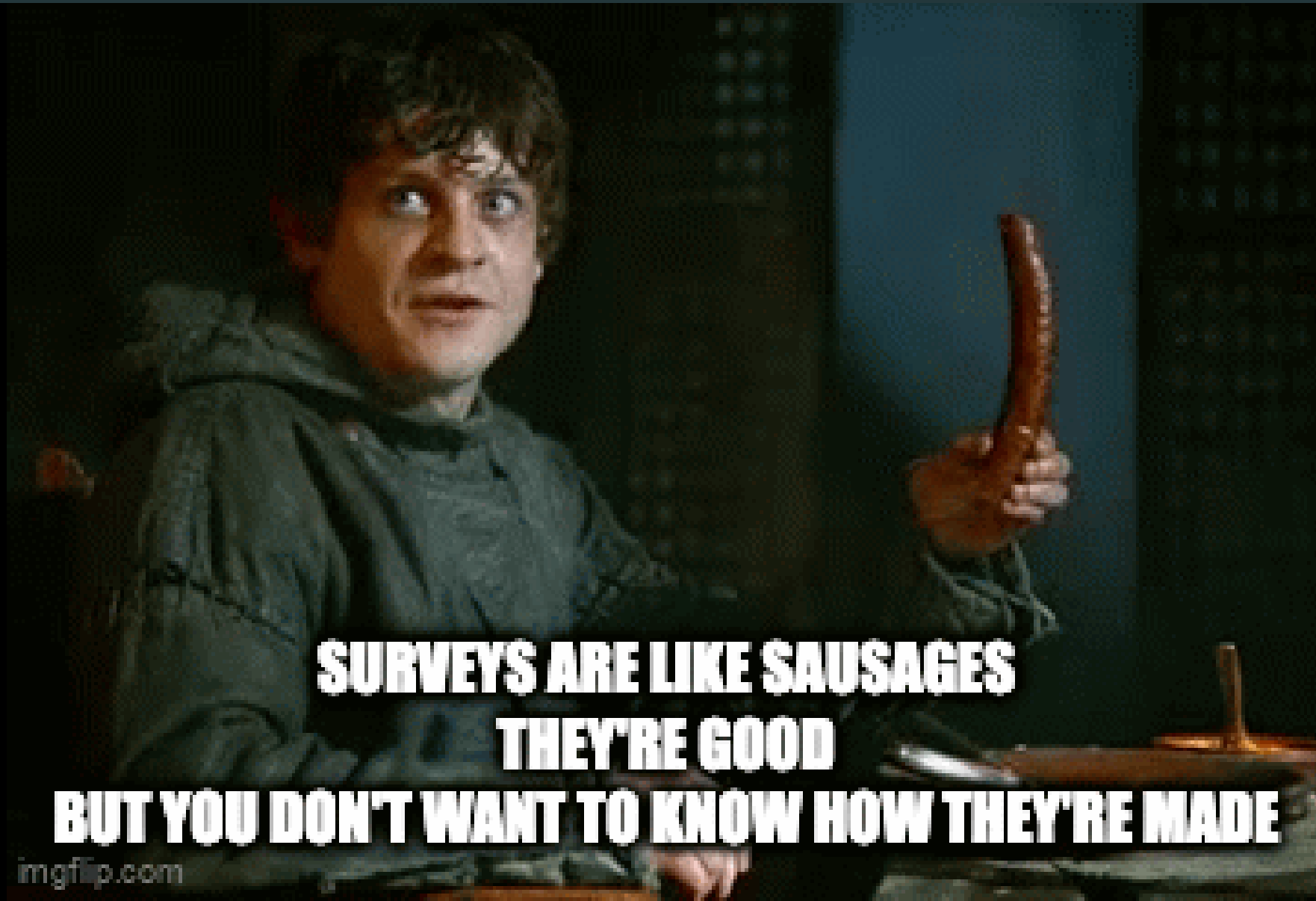
The Presserat found **no violation** — the editors had no reason to doubt their partner's claims. The ruling was widely criticized: the Presserat effectively declined to assess methodological quality.

Richtlinie 2.1 sets a transparency floor, but it does not require — or enable — the Presserat to evaluate whether a survey's *methods* justify its claims.

Article [Stefan Fries, Deutschlandfunk](#)

How surveys are made

A wisdom to start with



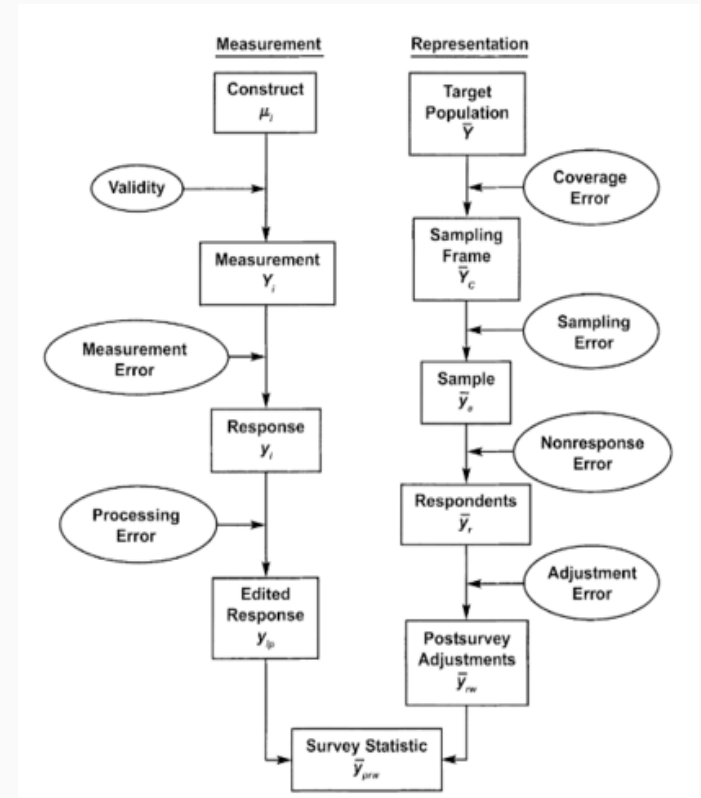
**SURVEYS ARE LIKE SAUSAGES
THEY'RE GOOD
BUT YOU DON'T WANT TO KNOW HOW THEY'RE MADE**

The survey production pipeline

From question to data

1. Define the research question — what do you want to learn?
2. Design the questionnaire — question wording, order, response scales
3. Select the survey mode — phone, online, face-to-face, mail
4. Draw the sample — from a sampling frame (register, panel, phone book)
5. Fieldwork — actually collect responses
6. Data processing — cleaning, coding, weighting
7. Analysis and reporting — estimates, margins of error, tables

Total Survey Error framework



Source Groves et al. 2009, Survey Methodology

The art of asking questions

Common pitfalls in questionnaire design

Leading questions: "Don't you agree that climate change is the most pressing issue?" → pushes respondents toward agreement.

Double-barreled questions: "Do you support increasing taxes and spending on education?" → conflates two separate opinions.

Acquiescence bias: respondents tend to agree with statements, regardless of content. Mitigated by mixing positively and negatively worded items.

Social desirability: respondents underreport socially stigmatized behavior (e.g., drug use, racist attitudes) and overreport socially approved behavior (voting, donating, exercising).

Classic question order experiment

"Do you think [the United States | a Communist country like Russia] should let [Communist | American] newspaper reporters from other countries come in here and send back to their papers the news as they see it?"

Table 1. Percent Yes to the Communist Reporter Question by Context and Contiguity

<i>Order</i>	<i>A Com./Amer.</i>	<i>B Amer./Com.</i>	<i>C Amer./ (17 items)/Com.</i>
Percent (Sample size)	44.4% (117)	70.1% (107)	66.4% (107)
	A vs. B: $\chi^2_1 = 15.20, p < 0.001^a$		
	B vs. C: $\chi^2_1 = 0.34, n.s.$		
	A vs. C: $\chi^2_1 = 10.95, p < 0.001^a$		

^a The overall χ^2 for the table is $\chi^2_2 = 18.11; p < 0.001$. For an approximate multiple comparisons test based on the Bonferroni inequality, the p -values may be increased by a factor of 3; the A vs. B contrast remains significant at the $p < 0.001$ level and the A vs. C contrast is significant at the $p < 0.005$ level.

Source [Schuman, Kalton and Ludwig 1983](#)

Measurement quality: a journalist's toolkit

What to watch for in question wording

Check for framing effects: was the question preceded by a vignette or preamble that could prime respondents? ("Given the recent terrorist attacks, do you support...")

Check for asymmetric scales: "Strongly agree / Agree / Neutral / Disagree" — where is "Strongly disagree"?
Unbalanced scales push responses toward one side.

Check for order effects: questions earlier in the survey can prime answers to later questions. If question 5 asks about crime, question 6 about immigration may get different answers than if the order were reversed.

Check for filter questions: did the survey screen out certain respondents before the key question? "Among those who said they are likely to vote..." — this subgroup may not resemble the full population.

Another question order experiment

The key belief question was embedded in the American Life Panel (ALP) and modeled after a survey by ABC News, Stanford University, and Time Magazine (2006). It read:

You may have heard about the idea that the world’s temperature may have been *going up* [*changing*] over the past 100 years, a phenomenon sometimes called ‘*global warming*’ [*climate change*]. What is your personal opinion regarding whether or not this has been happening? (Definitely **has not been** happening; Probably **has not been** happening; Unsure, but leaning toward it **has not been** happening; Not sure either way; Unsure, but leaning toward it **has been** happening; Probably **has been** happening; Definitely **has been** happening)³

Table 2. Distribution of Existence Beliefs by Political Self-identification and Question Wording (GW = “global warming”; CC = “climate change”)

Reported Existence Belief	Overall		Republicans		Democrats		Independents		Others	
	GW	CC	GW	CC	GW	CC	GW	CC	GW	CC
1 = Definitely has not been happening	6.6%	3.9%	12.7%	5.1%	1.9%	1.3%	6.6%	5.4%	4.7%	5.7%
2 = Probably has not been happening	8.4%	6.8%	18.2%	14.3%	2.6%	1.0%	6.2%	5.4%	3.7%	4.8%
3 = Leaning has not been happening	8.2%	7.6%	14.1%	11.4%	2.9%	5.2%	9.2%	6.6%	6.5%	5.7%
4 = Unsure	9.0%	7.6%	11.0%	8.9%	5.7%	6.0%	8.5%	8.7%	16.8%	6.7%
5 = Leaning has been happening	13.9%	15.7%	14.1%	17.0%	12.1%	13.6%	15.1%	15.7%	17.8%	18.1%
6 = Probably has been happening	25.6%	27.8%	20.2%	28.9%	29.2%	27.0%	27.9%	28.5%	23.4%	25.7%
7 = Definitely has been happening	28.2%	30.5%	9.7%	14.3%	45.6%	45.8%	26.5%	29.8%	27.1%	33.3%
% High Belief (≥ 5)	67.7%	74.0%	44.0%	60.2%	86.9%	86.4%	69.5%	74.0%	68.3%	77.1%
Mean Belief	5.05	5.30	3.95	4.62	5.94	5.94	5.09	5.29	5.18	5.37
N	1162	1099	362	370	421	382	272	242	107	105

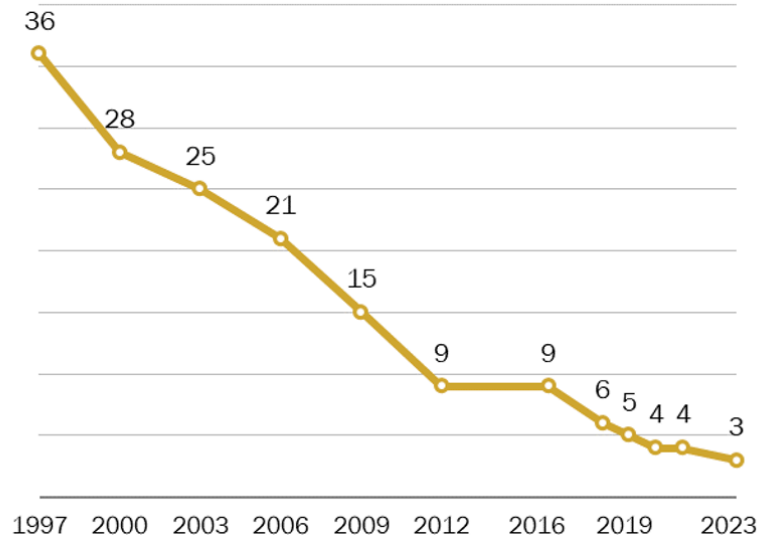
Source [Schuldt et al., POQ](#)

Survey modes: then and now

As phone response rates go down...

Telephone survey response rates, 1997-2023

% responding, based on AAPOR RR3



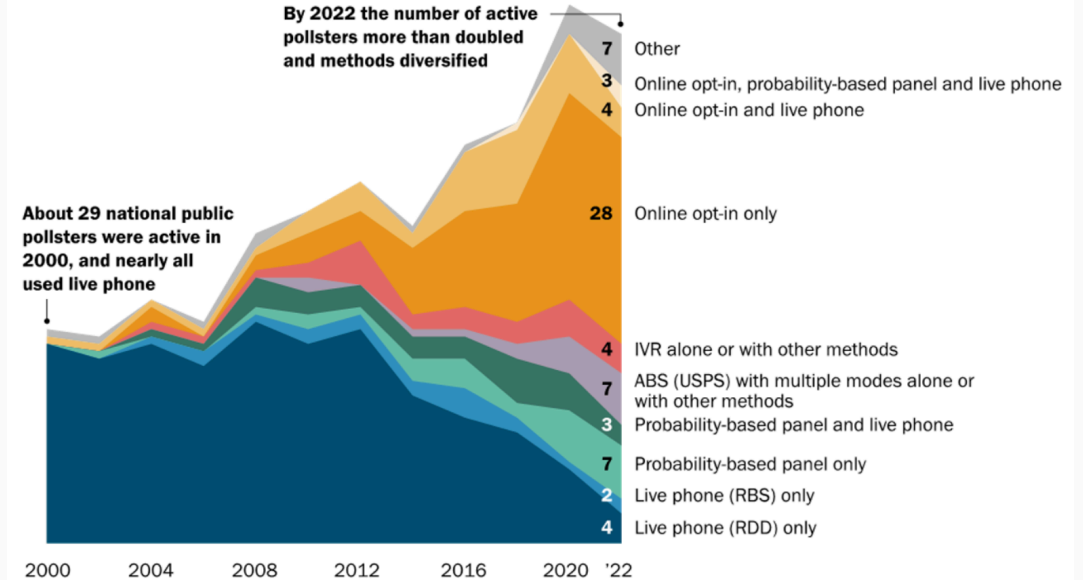
Annual averages of Pew Research Center telephone survey response rates

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...pollsters turn to online panels

Polling has entered a period of unprecedented diversity in methods

Number of national public pollsters in the U.S. using method(s)



Note: Figures represent the number of active national public pollsters in each year and the method(s) that they used. IVR refers to interactive voice response, also known as robo-polling. ABS refers to address-based sampling. RBS refers to voter registration-based sampling. RDD refers to random-digit-dial sampling.

Source: Pew Research Center analysis of external data. See Methodology for details. "How Public Polling Has Changed in the 21st Century"

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Survey modes: how data is collected

Traditional modes

CATI (Computer-Assisted Telephone Interview): random-digit dialing or registry-based. Still used by major pollsters (Infratest dimap, Forsa). Declining response rates are a serious problem — down to ~1–10% in many countries.

CAPI (Computer-Assisted Personal Interview): face-to-face, household-based. Gold standard for large social surveys (ESS, SOEP, ALLBUS). Expensive; used for long, complex questionnaires.

Mail / paper: still used for census and some government surveys. Slow but reaches populations without internet/phone.

Online modes

Access panels: pre-recruited pools of respondents who agreed to take surveys in exchange for incentives. Major providers: YouGov, Bilendi/Respondi, Dynata.

Crowdsourcing platforms: MTurk, Prolific, CloudResearch. Popular in academia; convenience samples with known demographic skews (younger, more educated, more liberal).

Self-selected / opt-in: anyone who feels like participating clicks a link. **Not a survey** in any rigorous sense — but often reported as one.

Access panels and the economics of survey production

How access panels work

1. A panel company (e.g., YouGov, Bilendi) **recruits** hundreds of thousands of people who agree to take surveys regularly.
2. For each new survey, a **sample is drawn from the panel** — either randomly or quota-based (e.g., targets for age × gender × region). 🚫 Even if drawn randomly from the panel, this is *not* a random sample of the population!
3. Panelists receive an **invitation** (email, app notification) and an **incentive** (points, vouchers, small payments).
4. The raw data is **weighted** to match known population benchmarks (from census or microcensus data).

Speed vs. quality trade-off

- A phone poll of $n = 1,000$ typically takes **3–5 days** of fieldwork.
- An online panel survey of $n = 2,000$ can be fielded in **24–48 hours**.
- Speed matters for tracking fast-moving opinion (elections, crises) but can sacrifice quality.

Cost per completed interview (approximate)

Mode	CPI
Face-to-face (CAPI)	€50–150
Phone (CATI)	€15–40
Online panel	€1–10
Opt-in / self-selected	~€0

Pew study (2023) on probability vs. opt-in samples

The study

- 6 parallel online surveys of ~5,000 U.S. adults each, benchmarked against 28 government data sources
- Opt-in samples had twice the error of probability panels (5.8 vs. 2.6 pp average absolute error)
- Opt-in errors worst for hard-to-reach groups: 11.2 pp for 18–29-year-olds, 10.8 pp for Hispanic adults — largely driven by ~8–15% "bogus respondents" who say Yes to everything

Example: Measuring error for a benchmark variable

	Survey estimate	Benchmark value	Error	Absolute error
Smoking status				
Smoke every day	15%	9%	+6	6
Smoke some days	8%	3%	+5	5
Do not now smoke	19%	22%	-3	3
Never smoked 100 cigarettes	55%	63%	-8	8

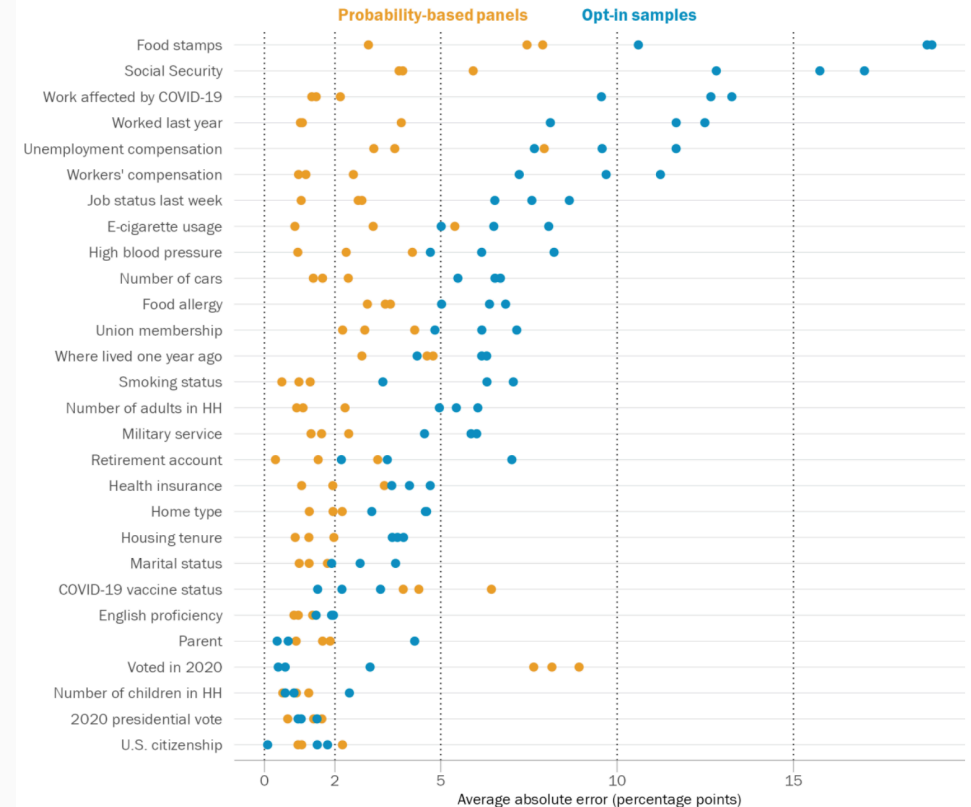
Average absolute error = $(6 + 5 + 3 + 8)/4 = 5.5$

Note: Benchmark values for smoking are from the 2020 National Health Interview Survey. The survey estimates shown were created as an example and do not correspond to any of the samples included in this study.

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Samples of the same type generally had similar levels of error on individual benchmarks

Average absolute error for estimates among U.S. adults



Source: Pew Research Center analysis of six online samples surveyed June 14-July 21, 2021.

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Source [Pew Research \(link to full report\)](#)

Survey sampling 101

Question: How would you define "representative survey"?

Klare Mehrheit der Deutschen für Investitionen in die Energiewende und gegen neue Gaskraftwerke

Montag, 01.09.2025

Dateien: 1

• **Repräsentative Umfrage** zeigt: 73 Prozent der wahlberechtigten Menschen finden die Energiewende wichtig für Deutschland

Hälfte der Deutschen will Verbot der Plattform X bei weiteren Rechtsverstößen

Viele Menschen sind mit ihrer Geduld mit Elon Musk und seiner Plattform X am Ende. Die Plattform hatte im Zusammenspiel mit dem Chatbot Grok zuletzt millionenfach sexualisierte Deepfakes verbreitet und ist seit der Übernahme durch den Milliardär politisch auf rechtsaußen gepolt.

11.02.2026 um 04:55 Uhr - Markus Reuter - in Öffentlichkeit - 14 Ergänzungen

Laut einer **repräsentativen Umfrage des Instituts YouGov** im Auftrag von People vs Big Tech und HateAid fordern knapp 70 Prozent der Deutschen, dass die EU mehr Maßnahmen ergreifen soll, wenn

Umfrage: Stabile Mehrheit von 65 % der Deutschen will Kernkraft

Veröffentlicht am 02.12.2024

Von Nuklearia

Eine seit drei Jahren laufende, **von Civey durchgeführte repräsentative Umfrage** belegt klar, dass Deutsche mit einer stabilen und deutlichen Mehrheit von ca. 65 % die Kernkraft befürworten. Gegen die Kernkraft sprechen sich weniger als 30 % der Befragten aus.

Seit Mai 2021 stellt Civey im Auftrag von Nuklearia e. V. kontinuierlich die Frage: »Sollte weiterhin Atomkraft eingesetzt werden, um die Klimaschutzziele der EU zu erreichen?« Das repräsentative Ergebnis zeigt: Die Befürworter der Kernkraft waren von Anfang an konstant in der Mehrheit. Mit Beginn der Energiekrise im September 2021 und dem Krieg in der Ukraine stieg die Zustimmung zur Kernenergie noch einmal stark an.

Seit Mitte 2022 liegen die Befürworter der Kernkraft mit mehr als 65 % deutlich vorn, während das Lager der Skeptiker ebenso gleichbleibend bei unter 30 % liegt. Die Stichprobengröße der Umfrage liegt bei über 10.000; der statistische Fehler im Gesamtergebnis beträgt lediglich 2,5%.

Robby @odor.bsky.social · 29d
Eine **repräsentative BUND-Umfrage** hat ergeben, dass 70 Prozent der Verbraucher*innen ein eindeutiges staatliches Tierhaltungskennzeichen wollen
www.bund.net/themen/aktue...



Mehrheit der Verbraucher*innen in Deutschland wollen klare Tierhaltungskennzeichnung
Eine repräsentative BUND-Umfrage hat ergeben, dass 70 Prozent der Verbraucher*innen ein eindeutiges staatliches Tierhaltungskennzeichen ...
© www.bund.net

coolmarx @coolmarx.bsky.social · 1mo
Mal ein kleiner Faktencheck für die bürgerliche Auseinandersetzung mit der DDR:

Über 86% der DDR Bevölkerung war für den (reformierten) Sozialismus, das ergab eine **repräsentative 1989 durchgeführte Umfrage**. Nur 5% der DDR Bevölkerung empfand einen kapitalistischen Weg für erstrebenswert.



In einer Umfrage von 1989 antworteten lediglich 5 % dass die DDR einen kapitalistischen Weg einschlagen sollte.

Quelle: Zentralinstitut für Jugendforschung (ZfJ), Leipzig, Meinungsbormeter November 1989

Studie: Eine pessimistische Jugend rückt nach rechts

Eine neue Studie stellt einen "deutlichen Rechtsruck" bei Jugendlichen in Deutschland fest. Als Ursache sehen Forscher Verunsicherung, fehlenden Optimismus und geringe Widerstandsfähigkeit gegen Krisen. Dies stärke den Hang zu autoritären Konzepten.

Von **BR24** BR24 Redaktion

Jugendforscher sehen bei der jungen Generation in Deutschland einen "deutlichen Rechtsruck". Grund dafür sei die Sorge um die Sicherung des Wohlstands und die damit verbundene hohe politische Unzufriedenheit, erklärte der Sozialwissenschaftler Klaus Hurrelmann bei der Vorstellung der Studie **"Jugend in Deutschland 2024"** (externer Link auf Bezahl-Inhalt). In der **repräsentativen Umfrage** wurden den Angaben zufolge im Januar und Februar bundesweit rund 2.000 junge Menschen zwischen 14 und 29 Jahren befragt.

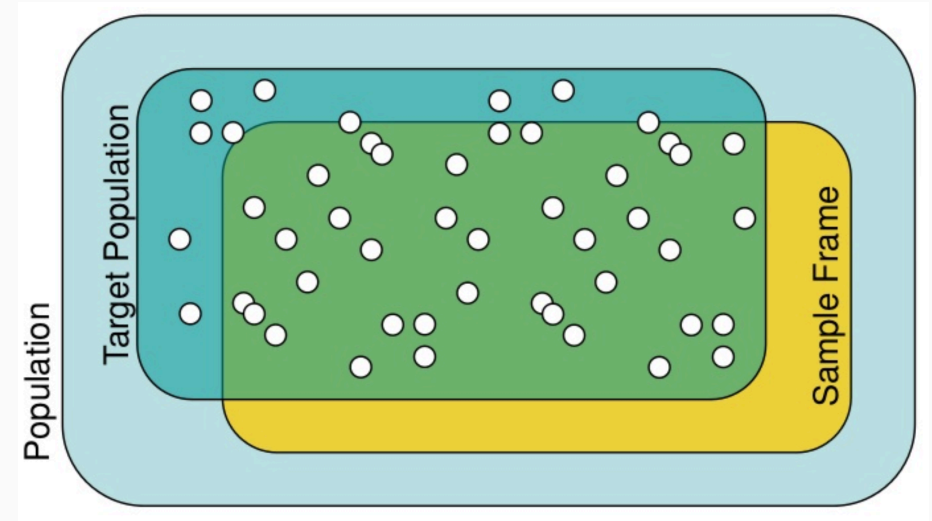
Representativity is in the eye of the beholder

A folk definition of representativity

A sample (or data in general) is "representative" if conclusions drawn from the sample can be generalized to the population of interest.

Why "representativity" is a problematic term

1. Whether a sample is representative **depends on your interest**. A sample can be representative for age and gender but not for income.
2. You cannot call a sample "representative" **a priori** — only after checking.
3. Assessing representativity requires **external benchmarks** (census, register data) — and those have their own limitations.



What this means for journalists

- Don't take reported "representativity" at face value.
- Sample size alone does not guarantee representativity.
- State specifically what the sample is supposed to be representative of (e.g., "representative of German adults aged 18+") and provide additional details on methods.
- Bad sampling is not restricted to surveys — think social media data, case selection for medical trials, or country selection for policy studies.

Why sampling (generally) works

The fundamental idea

If you draw a **random sample** from a population, the sample statistics (mean, proportion, etc.) will be close to the population parameters — and you can **quantify how close**.

This is the core insight of survey statistics and it rests on the **law of large numbers** and the **central limit theorem**.

Notation

- N = population size
- n = sample size
- p = true population proportion
- $\hat{p}$ = sample proportion
- $SE(\hat{p})$ = standard error of $\hat{p}$

The sampling distribution

If you drew many random samples of size n from a population and computed $\hat{p}$ each time, those estimates would form a **normal distribution** centered on p with standard deviation:

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

This gives us the **margin of error** at 95% confidence:

$$MoE = 1.96 \times SE(\hat{p})$$

Example

If $\hat{p} = 0.45$ and $n = 1,000$:

$$MoE = 1.96 \times \sqrt{\frac{0.45 \times 0.55}{1000}} \approx 0.031$$

So we could report: **45% $\pm$ 3.1 percentage points**.

How important is sample size?

The square root law

The margin of error shrinks with the **square root** of n :

$$\text{MoE} \propto \frac{1}{\sqrt{n}}$$

This has a sobering implication: **doubling accuracy requires quadrupling the sample size.**

n	MoE ($p = 0.5$)	MoE ($p = 0.25$)	MoE ($p = 0.1$)
100	±9.8%	±8.5%	±5.9%
400	±4.9%	±4.2%	±2.9%
1,000	±3.1%	±2.7%	±1.9%
2,000	±2.2%	±1.9%	±1.3%
10,000	±1.0%	±0.8%	±0.6%

Most national polls use $n = 1,000$ – $2,000$ because the gains beyond that are marginal relative to cost.

What sample size does NOT fix

Sample size controls **random error** (variance). It does **nothing** for **systematic error** (bias).

A biased sample of $n = 1,000,000$ is worse than an unbiased sample of $n = 1,000$.

"The 'bigness' of such Big Data (for population inferences) should be measured by the relative size $f = n/N$ of the sample to the population, not by the absolute size n of the sample."

Xiao-Li Meng (2018), *Annals of Applied Statistics*

👉 Stay tuned for the "Big Data Paradox" and the **R workshop script** `code/surveys/working-with-surveys.qmd` for a hands-on simulation.

Simple random sampling (SRS)

Definition

Every member of the population has an **equal probability** of being selected: $\pi_i = n/N$ for all i .

How it works in practice

1. Obtain a **sampling frame** — a list of all units in the population (e.g., voter registry, address register).
2. Assign each unit a random number.
3. Select the n units with smallest random numbers.

Variance of the sample mean under SRS

$$\text{Var}(\bar{y}_{SRS}) = \frac{S^2}{n} \left(1 - \frac{n}{N}\right)$$

where $S^2 = \frac{1}{N-1} \sum_{i=1}^N (y_i - \bar{Y})^2$ and the term $(1 - n/N)$ is the **finite population correction** (FPC).

Limitations

- Requires a complete sampling frame
- Can still yield unbalanced samples by chance
- Inefficient for rare subpopulations
- Sampling frame $\neq$ realized sample

The sampling frame problem

In practice, **no perfect sampling frame exists** for most populations. Phone registries miss mobile-only households. Address registers miss the homeless, recent migrants, and institutional populations.

Key insight

When a poll says "random sample," ask: **random from what list?** The frame determines who *can* be reached, not who ended up in the survey.

Stratified sampling

Definition

Divide the population into non-overlapping **strata** (e.g., by age group, region, education) and draw an independent random sample within each stratum.

Why stratify?

1. **Guarantees representation** of key subgroups.
2. **Reduces variance** — if strata are internally homogeneous, within-stratum variance is smaller than overall variance.
3. **Enables reliable subgroup estimates** — you can oversample small but important groups.

One can also oversample small/important strata (e.g., East German states). Requires **weighting** to produce unbiased population estimates.

Variance reduction

For a stratified sample with proportional allocation, the variance of the estimated mean is:

$$\text{Var}(\bar{y}_{st}) = \sum_{h=1}^H W_h^2 \frac{S_h^2}{n_h}$$

where $W_h = N_h/N$ is the stratum weight and S_h^2 is the within-stratum variance.

If strata are well-chosen (homogeneous within, heterogeneous between), this is **always** $\leq$ the variance under SRS.

Optimal (Neyman) allocation

Allocate more interviews to strata with higher variance:

$$n_h \propto N_h \cdot S_h$$

Cluster sampling

Definition

Divide the population into **clusters** (e.g., schools, districts), randomly select a subset of clusters, then survey all (or a sample of) units within selected clusters.

Why cluster?

- Only list of clusters not individuals needed
- **Cost-efficient** for F2F surveys: interviewers only visit selected areas
- Standard design for large household surveys

The cost: Design effect

Cluster sampling is **less efficient** than SRS or stratified sampling. Units within a cluster tend to be similar (e.g., students in the same school), so each additional unit within a cluster adds less new information.

Design effect example

DEFF quantifies the efficiency loss from clustering:

$$\text{DEFF} = 1 + (m - 1) \times \rho$$

where m is the average cluster size and ρ is the **intra-cluster correlation** (ICC) — the degree to which units within a cluster resemble each other. The **effective sample size** is:

$$n_{\text{eff}} = n / \text{DEFF}$$

Example

If $m = 20$ (20 students per school) and $\rho = 0.05$:

$$\text{DEFF} = 1 + 19 \times 0.05 = 1.95$$

So a cluster sample of $n = 2,000$ has the precision of an SRS of only $n \approx 1,026$.

Multi-stage sampling: what real surveys look like

Combining strategies

Most real-world surveys use **multi-stage** designs that combine stratification and clustering:

Stage 1: Stratify by region (e.g., 16 Bundesländer) and urbanization (urban/rural).

Stage 2: Within each stratum, randomly select **clusters** (e.g., municipalities or sampling points).

Stage 3: Within selected clusters, randomly select **households** (from address registers).

Stage 4: Within selected households, randomly select **one person** (e.g., using the Kish grid or last-birthday method).

This is how the ALLBUS, ESS, SOEP, and most major social surveys work.

Why this matters for analysis

Standard formulas (assuming SRS) **underestimate standard errors**. Analysts must account for:

- **Stratification** (reduces SE)
- **Clustering** (increases SE)
- **Unequal selection probabilities** (requires weighting)

The R `survey` package handles all of this — you specify the design, and it computes correct standard errors.

```
library(survey)
des <- svydesign(
  ids = ~cluster_id,      # clustering
  strata = ~stratum,      # stratification
  weights = ~weight,      # design weights
  data = my_data
)
svymean(~outcome, des) # correct SE
```

Weighting: fixing what sampling broke

Why weight?

Even with probability sampling, the realized sample rarely matches the population perfectly. Weighting adjusts for:

1. **Unequal selection probabilities** — if you oversampled East Germany, downweight East Germans.
2. **Nonresponse** — if young men are underrepresented, upweight those who did respond.

How it works

Each respondent i receives a weight w_i .

$$\hat{p}_w = \frac{\sum_i w_i \cdot y_i}{\sum_i w_i}$$

where $y_i = 1$ if resp. i has the attribute of interest.

Common weighting approaches

Post-stratification: partition the sample into cross-classified cells (e.g., age × gender × region) and adjust weights so each cell matches its known population share from census data.

Raking / iterative proportional fitting: iteratively adjust weights to match multiple marginal distributions simultaneously — useful when the full cross-classification is unavailable or has too many empty cells.

Limitations

Weighting cannot fix **fundamental coverage bias** — if a group is missing from the sample, no weight can recover them.

Extreme weights increase variance (a few respondents dominate). Good practice: trim weights at 4–5× the mean.

Weighting on the **wrong variables** can make things worse.

Weighting in practice: an example

Scenario

You conducted an online panel survey (n = 1,000) on attitudes toward a four-day work week. Your sample is too young and too educated compared to the German population.

Group	Census %	Sample %
18–29	15%	28%
30–49	30%	35%
50–64	28%	25%
65+	27%	12%

Post-stratification weights

$$w_i = \frac{\text{Census proportion for group}_i}{\text{Sample proportion for group}_i}$$

For a 65+ respondent: $w = 27/12 = 2.25$. Their response counts 2.25× as much.

Effect on estimates

Suppose support for the four-day work week varies by age:

Group	Support	Unweighted share	Weighted share
18–29	72%	28% → overrepresented	15%
30–49	58%	35% → slight over	30%
50–64	41%	25% → slight under	28%
65+	25%	12% → underrepresented	27%

Unweighted estimate:

$$0.28 \times 72 + 0.35 \times 58 + 0.25 \times 41 + 0.12 \times 25 = 53.7\%$$

Weighted estimate:

$$0.15 \times 72 + 0.30 \times 58 + 0.28 \times 41 + 0.27 \times 25 = 46.4\%$$

A 7 percentage point difference — the unweighted number was misleading because it overrepresented the pro-reform age group.

A modern big data polling disaster

Unrepresentative big surveys significantly overestimated US vaccine uptake

<https://doi.org/10.1038/s41586-021-04198-4>

Received: 18 June 2021

Accepted: 29 October 2021

Published online: 8 December 2021

 Check for updates

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Surveys are a crucial tool for understanding public opinion and behaviour, and their accuracy depends on maintaining statistical representativeness of their target populations by minimizing biases from all sources. **Increasing data size shrinks confidence intervals but magnifies the effect of survey bias: an instance of the Big Data Paradox¹.** Here we demonstrate this paradox in estimates of first-dose COVID-19 vaccine uptake in US adults from 9 January to 19 May 2021 from two large surveys: Delphi–Facebook^{2,3} (about 250,000 responses per week) and Census Household Pulse⁴ (about 75,000 every two weeks). In May 2021, Delphi–Facebook overestimated uptake by 17 percentage points (14–20 percentage points with 5% benchmark imprecision) and Census Household Pulse by 14 (11–17 percentage points with 5% benchmark imprecision), compared to a retroactively updated benchmark the Centers for Disease Control and Prevention published on 26 May 2021. Moreover, their large sample sizes led to miniscule margins of error on the incorrect estimates. By contrast, an Axios–Ipsos online panel⁵ with about 1,000 responses per week following survey research best practices⁶ provided reliable estimates and uncertainty quantification. We decompose observed error using a recent analytic framework¹ to explain the inaccuracy in the three surveys. We then analyse the implications for vaccine hesitancy and willingness. **We show how a survey of 250,000 respondents can produce an estimate of the population mean that is no more accurate than an estimate from a simple random sample of size 10.** Our central message is that data quality matters more than data quantity, and that compensating the former with the latter is a mathematically provable losing proposition.

Source [Bradley et al. \(2021\), Nature](#)

Table 1 | Comparison of survey designs

	Axios-Ipsos	Census Household Pulse	Delphi-Facebook
Recruitment mode	Address-based mail sample to Ipsos KnowledgePanel	SMS and email	Facebook Newsfeed
Interview mode	Online	Online	Online
Average size	1,000/wave	75,000/wave	250,000/week
Sampling frame	Ipsos KnowledgePanel; internet/tablets provided to ~5% of panelists who lack home internet	Census Bureau's Master Address File (individuals for whom email / phone contact information is available)	Facebook active users
Vaccine uptake question	"Do you personally know anyone who has already received the COVID-19 vaccine?"	"Have you received a COVID-19 vaccine?"	"Have you had a COVID-19 vaccination?"
Vaccine uptake definition	"Yes, I have received the vaccine"	"Yes"	"Yes"
Other vaccine uptake response options	"Yes, a member of my immediate family", "Yes, someone else", "No"	"No"	"No", "I don't know"
Weighting variables	Gender by age, race, education, Census region, metropolitan status, household income, partisanship.	Education by age by sex by state, race/ethnicity by age by sex by state, household size	Stage 1: age, gender "other attributes which we have found in the past to correlate with survey outcomes" to FAUB; Stage 2: state by age by gender

Comparison of key design choices across the Axios–Ipsos, Census Household Pulse and Delphi–Facebook studies. All surveys target the US adult population. See Extended Data Table 1 for additional comparisons and Methods for additional implementation details.

A modern big data polling disaster

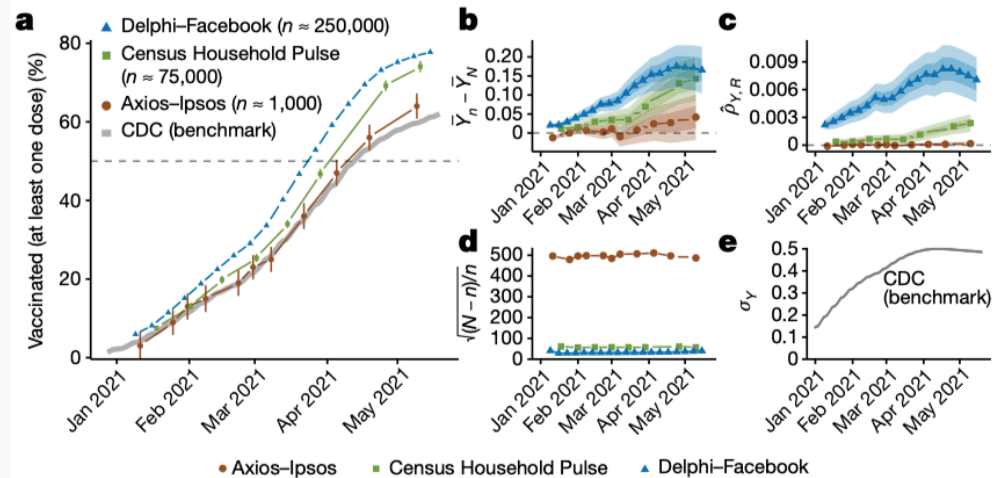


Fig1 | Errors in estimates of vaccine uptake.
a, Estimates of vaccine uptake for US adults in 2021 compared to CDC benchmark data, plotted by the end date of each survey wave. Points indicate each study's weighted estimate of first-dose vaccine uptake, and intervals are 95% confidence intervals using reported standard errors and design effects. Delphi-Facebook has $n = 4,525,633$ across 19 waves, Census Household Pulse has $n = 606,615$ across 8 waves and Axios-Ipsos has $n = 11,421$ across 11 waves. Delphi-Facebook's confidence intervals are too small to be visible. **b**, Total error $\bar{Y}_n - \bar{Y}_N$. **c**, Data defect correlation $\hat{\rho}_{Y,R}$. **d**, Data scarcity $\sqrt{(N-n)/n}$. **e**, Inherent problem difficulty σ_Y . Shaded bands represent scenarios of $\pm 5\%$ (darker) and $\pm 10\%$ (lighter) imprecision in the CDC benchmark relative to reported values (points). **b-e** comprise the decomposition in equation (1).

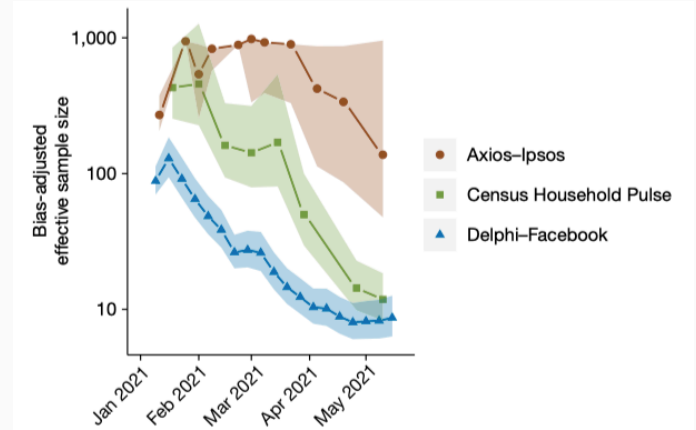


Fig 2 | Bias-adjusted effective sample size. An estimate's bias-adjusted effective sample size (different from the classic Kish effective sample size) is the size of a simple random sample that would have the same MSE as the observed estimate. Effective sample sizes are shown here on the log₁₀ scale. The original sample size was $n = 4,525,633$ across 19 waves for Delphi-Facebook, $n = 606,615$ across 8 waves for Census Household Pulse and $n = 11,421$ across 11 waves for Axios-Ipsos. Shaded bands represent scenarios of $\pm 5\%$ benchmark imprecision in the CDC benchmark.

The big data paradox: formal statement

"[W]hen biased samples are large, they are doubly misleading: they produce confidence intervals with incorrect centres and substantially underestimated widths. This is the **Big Data Paradox**: the bigger the data, the surer we fool ourselves when we fail to account for bias in data collection."

Bradley et al. (2021), *Nature*

The key insight

The **total error** of an estimate from a non-probability sample is:

$$\text{Total error} = \underbrace{\text{Bias}}_{\text{systematic}} + \underbrace{\text{SE}}_{\text{random}}$$

As (n) grows, SE shrinks — but **bias stays constant** (or can even grow if larger samples attract different types of respondents). The confidence interval narrows around the **wrong value**.

The Meng decomposition

Xiao-Li Meng (2018) showed that the error of a non-probability estimate can be decomposed into:

$$\hat{\bar{Y}} - \bar{Y} = \rho_{R,Y} \times \sigma_Y \times \sqrt{\frac{N-n}{n}}$$

where $\rho_{R,Y}$ is the correlation between being in the sample and the outcome — the **data defect index**.

Even tiny $\rho_{R,Y}$ (e.g., 0.005) can produce massive errors in large samples.

What this means

- A Facebook survey with $n = 250,000$ and $\rho_{R,Y} = 0.004$ can have the **same total error** as an SRS of $n = 10$.
- The standard error formula $\sqrt{p(1-p)/n}$ dramatically underestimates uncertainty for non-probability samples.
- **Size is not a substitute for randomization.**

Puzzle: What's going on here? (Selb and Munzert 2013)

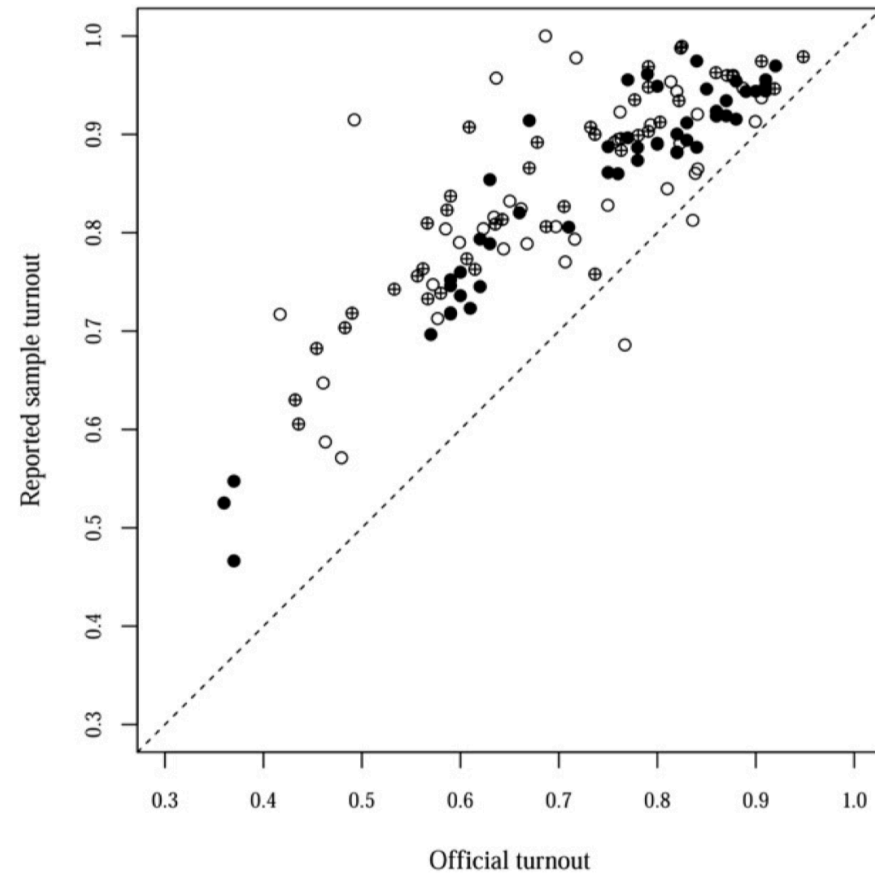


Fig. 1. Reported sample turnout rates from 130 postelection surveys versus official turnout. Data are taken from Modules 1–3 of the *Comparative Study of Electoral Systems* (CSES), and from a collection of election surveys for which vote validation studies (VVS) are available. For more detailed information, see the [Appendix](#) to this paper.

Take-aways

Take surveys with a pinch of salt



Session takeaways

Key lessons

1. **Sample size matters — but bias matters more.** A biased sample of $n = 1M$ is worse than an unbiased sample of $n = 1,000$.
2. **"Representative" is not a magic word.** Always ask: representative of what? Verified how?
3. **The margin of error only captures random error.** Systematic errors (coverage, nonresponse, measurement) are not included.
4. **Weighting helps — but has limits.** It can adjust for known imbalances but cannot fix coverage gaps.
5. **Question wording shapes answers.** Always check the exact wording before reporting.
6. **MRP is powerful but opaque.** Know that the poll you're reporting may be partly a model.

What a data journalist should do

- Always look for the **methodology statement** before reporting survey results.
- Report the **margin of error** and sample size alongside any point estimate.
- Be cautious with **subgroup estimates** — check the subgroup n .
- Be skeptical of **opt-in surveys** presented as representative polls.
- Use **multiple sources** — if three independent polls agree, the signal is stronger.
- Understand the basics of **weighting and MRP** so you can evaluate modern polling methods.

Don't contribute to this...

Transforming Stability into Change: How the Media Select and Report Opinion Polls

The International Journal of Press/Politics
2020, Vol. 25(1) 115–134
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DOI: 10.1177/1940161219864295
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Abstract

Although political polls show stability over short periods of time, most media coverage of polls highlights recurrent changes in the political competition. We present evidence for a snowball effect where small and insignificant changes in polls end up in the media coverage as stories about changes. To demonstrate this process, we rely on the full population of political polls in Denmark and a combination of human coding and supervised machine learning of more than four thousand news articles. Through these steps, we show how a horserace coverage of polls about change can rest on a foundation of stability.

Resources

Books

- **Groves et al. (2009)**: *Survey Methodology*, 2nd edition. The comprehensive reference — chapters on sampling, nonresponse, and measurement are excellent.
- **Lohr (2022)**: *Sampling: Design and Analysis*, 3rd edition. More technical; great for sampling design and estimation.
- **Wolf & Joye (2016)**: *The SAGE Handbook of Survey Methodology*. Good overview chapters on total survey error, modes, and questionnaire design.

Online resources

- [AAPOR Transparency Initiative](#) — standards for survey disclosure.
- [Pew Research Methods](#) — detailed methodology reports.
- [The Economist's MRP model](#) — explanation of their election forecasting approach.
- [Andrew Gelman's blog](#) — commentary on polling methodology and MRP.
- [Lauren Kennedy & Jonah Gabry: MRP primer](#) — accessible tutorial on MRP with rstanarm.

R packages

- `survey` / `srvyr` — analyzing complex survey data (design effects, weights, stratification).
- `anesrake` — raking/calibration weights.
- `WeightIt` — unified interface for survey weighting methods.